Vol 9 Chapter 9 — Photonic Crystals: The $10K BST Falsifier

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

Chapter 9 — Photonic Crystals: The $10K BST Falsifier

Level 1 — one sentence

Photonic crystals — periodic dielectric structures with optical band gaps analogous to electronic band gaps in solids — provide BST's cheapest experimental test: a specifically-engineered photonic crystal with BST-primary-anchored lattice parameters is predicted to exhibit a substrate eigentone resonance observable in standard transmission/reflection spectroscopy, total cost ~$10K.

Level 2 — graduate-physicist precision

9.1 Photonic crystals: the optical analog of band structure

A photonic crystal is a periodic dielectric structure: alternating layers (1D), patterned 2D slab, or 3D opal/inverse-opal structure with periodic refractive index variation.

By analogy with electronic band theory: - Bloch's theorem for EM waves - Photonic bands $\omega_n(\vec{k})$ - Photonic band gaps (frequencies forbidden in any direction) - Surface modes, defect states

Yablonovitch 1987: theoretical proposal of complete photonic band gaps. Experimental realization in 3D structures by 1991. Now standard in photonics engineering.

Applications: low-loss waveguides, light-trapping for solar cells, control of spontaneous emission rate, photonic-crystal lasers.

9.2 The BST eigentone prediction

Prediction: A photonic crystal with periodicities chosen as BST primary integer ratios — specifically, structures with characteristic lengths ℓ_1, ℓ_2, ℓ_3 in ratio $N_c : n_C : g = 3 : 5 : 7$ or with overall scale anchored to $N_{\max} = 137$ wavelengths per characteristic length — predicted to exhibit a substrate-natural eigentone at a frequency derivable from BST primaries.

Observable: enhanced absorption / dispersion feature in optical transmission spectrum of BST-engineered photonic crystal vs control structures with non-BST periodicity ratios.

9.3 Experimental design

Apparatus: - BST-engineered photonic crystal (3D opal or 2D slab; ~$1K fabrication via standard nanofab) - Control photonic crystal with non-BST ratios (also ~$1K) - Optical spectrometer (visible to mid-IR; standard equipment)

Protocol: - Measure transmission/reflection spectra on both samples - Look for substrate eigentone feature in BST sample absent in control - Verify predicted frequency matches BST-derived f_{phot}

Budget: ~$10K total (sample fabrication + spectrometer time + analysis). 3-6 month timeline.

Sites: any optics lab. Many universities and industry photonics groups.

9.4 Why even cheaper than $BaTiO_3$

The $BaTiO_3$ 137-plane test (Ch 8, ~$25K) requires acquiring specific-orientation single crystals + microwave/THz spectrometer access. Photonic crystals can be fabricated cheaply via standard nanofabrication; optical spectroscopy is more widely accessible than THz.

This makes the photonic-crystal test BST's cheapest experimental falsifier — accessible to most photonics groups worldwide.

9.5 Status

SP-30 parent program (BST task #185); photonic crystal specific design as sub-task pending Casey send-signal for outreach.

9.6 Other photonic-crystal applications

Beyond BST falsifier, photonic crystals have many practical uses: - Photonic bandgap fibers (low-loss for telecommunications) - Slow-light structures (group velocity → 0 near band edges) - Lasing in photonic-bandgap defects (mode confinement) - Solar cells (light trapping via photonic-crystal back reflector) - Optical metamaterials (negative-index, cloaking, perfect lensing)

9.7 K-audit anchors

- Vol 0 Ch 5: $N_{\text{max}} = 137$ + BST primaries (basis for eigentone prediction)
- SP-30 task #185 (substrate engineering)
- Vol 7 Ch 10: EM in matter (foundation for photonic crystals)

Level 3 — 5th-grader accessibility

Cheapest BST test ever: build a photonic crystal (periodic optical structure, ~$1K fab cost). Engineer the periodicities to match BST primary integer ratios ($3 : 5 : 7$, or scaled by $N_{\text{max}} = 137$). BST predicts you'll see a specific substrate eigentone resonance in the optical transmission spectrum. Total cost ~$10K (sample + spectroscopy + analysis). 3-6 months. Any optics lab can do it.

- Eigentone observed at predicted frequency → BST confirmed
- Not observed → BST substrate-mechanism for photonics is wrong

The cheapest go/no-go test of BST in all of physics. Cheaper than a year of grad-student funding.

What comes next

Chapter 10 develops strongly correlated systems.

Where to look this up

- Joannopoulos et al., *Photonic Crystals: Molding the Flow of Light*
- BST: Vol 0 Ch 5, SP-30 task #185